

Lovely Professional University

School of Computer Science and Engineering

UpGrad Campus, Phagwara, Punjab – India



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SYNOPSIS

(for the B.Tech Final Year Project)

AI Agent for Personalized Treatment Recommendation

(An Intelligent Multi-Agent System for Disease Prediction and Personalized Healthcare Guidance using Machine Learning and Generative AI)

Submitted by:

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Course Code: CSM355

Session: Jan 2026 – May 2026

Introduction:

Background/Context:

With the rapid advancement in digital healthcare and artificial intelligence, there is an increasing demand for intelligent systems that can assist individuals in understanding their health conditions and receiving timely medical guidance. Traditional healthcare systems often rely on manual diagnosis and generalized treatment procedures, which may not always consider individual patient differences such as medical history, lifestyle, and symptoms.

Machine Learning (ML) and Generative Artificial Intelligence (GenAI) have introduced new possibilities in healthcare by enabling data-driven diagnosis, automated report analysis, and personalized treatment recommendations. These technologies can analyze large volumes of structured and unstructured medical data, including patient symptoms and clinical reports, to identify patterns and generate meaningful insights.

This project focuses on developing an AI-powered agent that integrates ML, NLP, and GenAI to provide accurate disease prediction and personalized treatment recommendations, thereby improving accessibility and efficiency in healthcare support systems.

Problem Statement:

Many individuals face difficulty in obtaining quick and reliable preliminary medical advice, especially in remote or resource-limited areas. Existing systems either provide generic suggestions or lack the capability to analyze detailed medical reports and patient-specific data.

Moreover, traditional healthcare applications do not effectively combine machine learning predictions with intelligent reasoning and contextual understanding. This leads to limitations in personalization and accuracy.

This project aims to address these challenges by developing an AI-based system that can analyze patient symptoms and medical reports, predict possible diseases, and generate personalized treatment recommendations using advanced ML and GenAI techniques.

Rationale/Justification:

The increasing need for digital healthcare solutions and decision support systems makes this project highly relevant. By combining **Machine Learning, Natural**

Language Processing(NLP), and Generative AI, the system can provide intelligent, scalable, and personalized healthcare assistance.

This project not only contributes to improving healthcare accessibility but also demonstrates the practical application of AI technologies in solving real-world problems. It aligns with current trends in AI-driven healthcare innovation and provides a strong foundation for future research and development.

Objectives:

General Objective:

To design and implement an AI-powered system that provides personalized treatment recommendations based on patient data using machine learning and generative AI techniques.

Specific Objectives:

- To collect and preprocess healthcare datasets containing symptoms, diseases, and medical records.
 - To develop machine learning models (XGBoost, Random Forest) for accurate disease prediction.
 - To implement natural language processing techniques for extracting information from medical reports.
 - To build a Generative AI-based agent capable of reasoning and generating personalized treatment suggestions.
 - To integrate a Retrieval-Augmented Generation (RAG) system for accessing reliable medical knowledge.
 - To design a user-friendly web interface for interacting with the AI system.
 - To include features such as risk assessment, prevention strategies, and health recommendations.
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Scope of the Project:

Included:

- Development of a disease prediction system using machine learning algorithms.
- Implementation of medical report analysis using OCR and NLP techniques.
- Integration of a generative AI agent for personalized treatment recommendations.
- Design of a web-based application for user interaction.

- Inclusion of features such as symptom input, report upload, and health guidance.
 - Evaluation of system performance using standard machine learning metrics.
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Excluded:

- Real-time integration with hospital databases or electronic health records (EHR).
 - Direct prescription of medications without professional medical validation.
 - Integration with wearable devices or real-time health monitoring systems.
 - Emergency or critical medical decision-making applications.
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Literature Review (brief):

- **Artificial Intelligence in Healthcare:** AI has been widely used for disease prediction, diagnosis support, and healthcare automation.
 - **Machine Learning Models in Medicine:** Algorithms such as XGBoost and Random Forest have shown high accuracy in classification tasks related to disease prediction.
 - **Natural Language Processing in Healthcare:** NLP techniques are used to extract meaningful information from clinical text and medical reports.
 - **Generative AI Applications:** Large Language Models (LLMs) are increasingly used for intelligent chatbots, recommendation systems, and decision support in healthcare.
 - **Clinical Decision Support Systems:** AI-based systems improve diagnostic accuracy and assist healthcare professionals in making informed decisions.
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Methodology:**Research Design:**

The project follows a quantitative and experimental approach using healthcare datasets and AI models for prediction and recommendation.

Data Collection Methods:

- Publicly available healthcare datasets from platforms such as Kaggle and UCI Machine Learning Repository.
- Sample medical reports for testing report analysis modules.

- Structured and unstructured patient data including symptoms and diagnostic parameters.
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Data Analysis Techniques:

- Feature engineering on healthcare datasets to improve model performance.
 - Disease prediction using machine learning models such as XGBoost and Random Forest.
 - Text extraction and processing using NLP techniques (spaCy).
 - Report analysis using OCR tools like Tesseract.
 - Implementation of Generative AI models for reasoning and recommendation generation.
 - Use of Retrieval-Augmented Generation (RAG) for improving accuracy and reducing hallucination.
 - Evaluation using metrics such as accuracy, precision, recall, and F1-score.
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Tools and Technologies:

- **Languages:** Python
 - **Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, spaCy
 - **GenAI:** OpenAI GPT / LLaMA
 - **OCR:** Tesseract
 - **Backend:** FastAPI
 - **Frontend:** Streamlit
 - **Database:** FAISS / Chroma (Vector Database)
 - **Visualization:** Matplotlib, Plotly
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Expected Outcomes/Deliverables:

- A web-based AI system for disease prediction and personalized treatment recommendation.
- A functional module for analyzing medical reports and extracting key insights.
- An intelligent AI agent capable of multi-step reasoning and decision-making.
- A user-friendly interface for interacting with the system.
- A comparative analysis of machine learning models used for prediction.
- A demonstration of integration between ML, NLP, and Generative AI technologies.
- Contribution to healthcare decision-support systems and AI research.

Project Timeline:

Phase	Activity
Week 1–2	Data Collection and Preprocessing
Week 3–4	Machine Learning Model Development
Week 5–6	NLP and Report Analysis
Week 7	GenAI Integration
Week 8	RAG Implementation
Week 9	UI Development and Testing
Week 10	Final Report and Submission

References (APA Style):

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 2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python.
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 6. OpenAI. (2023). GPT models and applications.
 7. Jurafsky, D., & Martin, J. H. (2020). Speech and Language Processing.
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Declaration:

I hereby declare that the research work titled “**AI Agent for Personalized Treatment Recommendation**” is an original work carried out by me in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering.

I affirm that this work has not been submitted elsewhere for any degree or diploma. All sources of information have been duly acknowledged, and the content reflects my independent research and development effort under the guidance of my supervisor **Mr. Hritwiz Yash**.

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Signatures

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Supervisor:

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